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Tanks, Vessels, and Valves

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Introduction

- Tanks, vessels, and valves are vital components in chemical plants.
- They store, mix, and control the flow of liquids and gases safely.
- Pressure and temperatures are maintained in tanks and vessels, and valves are used to control the flow and direction in pipelines.
- This is done together to provide a smooth, efficient and safe operation of plants.

Importance of Equipment

Equipment enables conversion, separation, and transport of materials.

Correct design ensures safety, efficiency, and product quality.

Equipment selection affects cost, energy use, and sustainability.

Knowing equipment helps engineers scale up processes from lab to plant.

Categories of Equipment

Category	Purpose
Heat Exchangers, Furnaces, and Boilers	Energy transfer
Distillation Columns and Separators	Mixture separation
Pumps and Compressors	Fluid transport and pressure control
Reactors	Chemical transformation
Solid Handling Equipment	Filtration, drying, conveying
Tanks, Vessels, and Valves	Storage and flow regulation

Types of Storage Tanks

The Petro Solutions. (n.d.). *Types of storage tanks [Image].* Retrieved November 2, 2025, from https://thepetrosolutions.com/types-of-storage-tanks/

Fixed-Roof Tanks:
Cylindrical tanks with a permanent cone or dome roof, used for storing non-volatile liquids such as water or fuel oil.

External Floating-Roof (EFR) Tanks:
Roof floats directly on the liquid surface, minimizing vapor losses in volatile hydrocarbons like gasoline and crude oil.

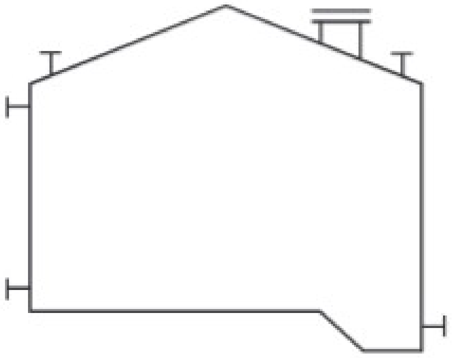
Internal Floating-Roof (IFR) Tanks:
Have a fixed outer roof with a floating internal deck to control emissions and prevent contamination.

Horizontal and Rectangular Tanks:
Used for small-volume or auxiliary storage when vertical height is limited, often for chemicals, fuels, or water.

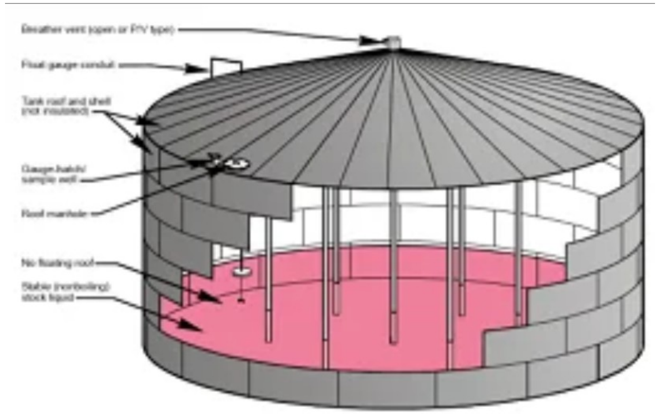
LNG Storage Tanks:
Double-walled, insulated tanks designed to store cryogenic liquids such as liquefied natural gas at extremely low temperatures.

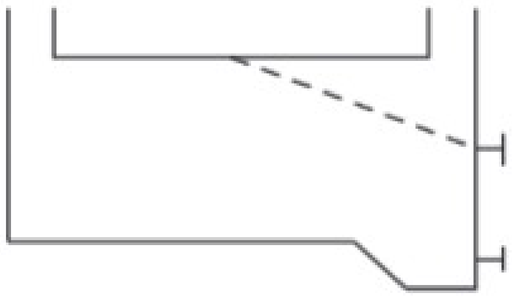
Spherical Storage Tanks:
Spherical vessels used for high-pressure liquids (e.g., propane, butane, ammonia).

Storage Tank Diagrams

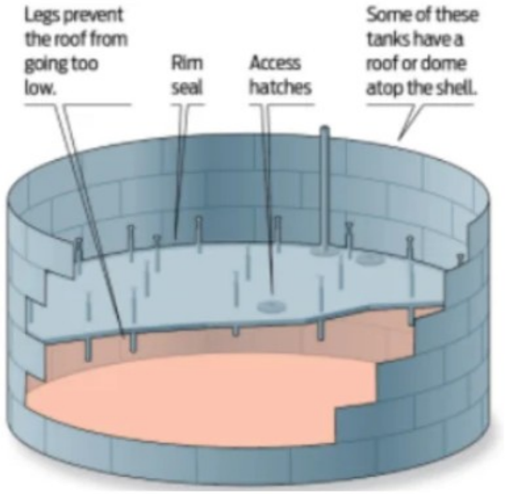


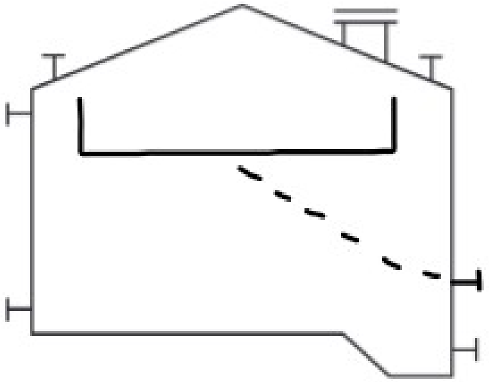
Fixed-Roof Tanks



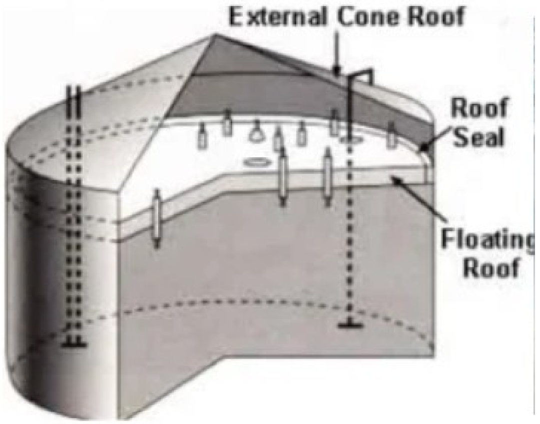


External Floating-Roof Tanks



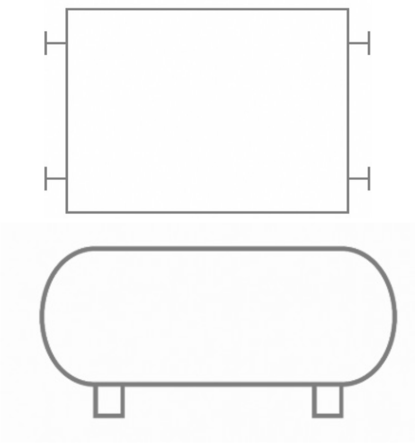


Internal Floating-Roof Tanks

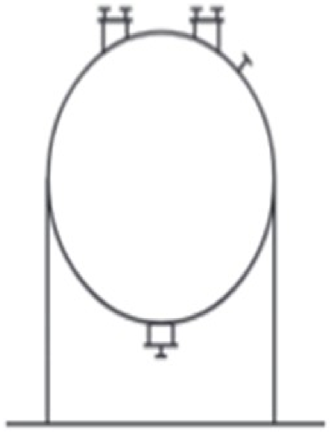


PFD symbols were either sourced from Gavin’s textbook or drawn by hand. Images are from source 2.

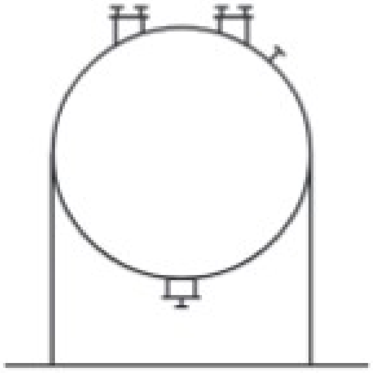
Storage Tank Diagrams - cont.



Horizontal and Rectangular Tanks



LNG Storage Tanks



Spherical Tanks

PFD symbols were either sourced from Gavin’s textbook or drawn by hand. Images are from source 2.

Storage Tanks Applications and Design Rules

Applications

- Used throughout chemical, petroleum, and energy industries for bulk liquid and gas storage.
- Found in refineries, petrochemical, and chemical plants for feed, product, and intermediate storage.
- Fixed-roof tanks store non-volatile liquids; floating-roof tanks handle volatile hydrocarbons.
- Horizontal and rectangular tanks are used for small-volume or auxiliary storage.
- LNG and spherical tanks store cryogenic or pressurized gases such as LNG, LPG, and ammonia.

Rules of Thumb

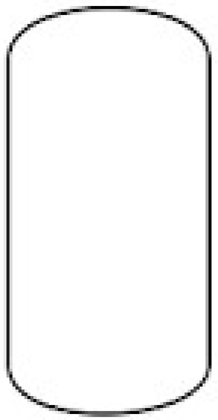
- Plate thickness increases from top to bottom to handle hydrostatic pressure in large tanks.
- Roof support (steel columns or framework) is needed for very large-diameter tanks.
- Use corrosion-resistant materials or coatings for long-term liquid containment.
- Allow vapor space (≈10–15%) above the liquid to accommodate expansion.
- Use vertical tanks for large volumes and horizontal tanks for smaller or height-limited installations.

Types of Vessels

Vertical	Typically, cylindrical vessels used for columns, reactors, tanks, and storage. Require more stabilization than other types of vessels and can be difficult to clean.
Horizontal	Typically, cylindrical vessels used for storage and separations with a larger footprint than vertical vessels.
Spherical	Used for storage of high-pressure gases; more costly to produce than other vessel types

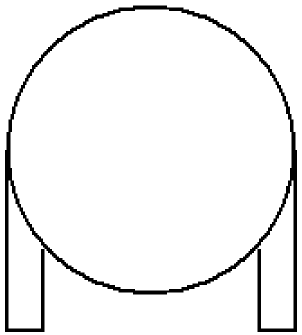
Images source: https://pressurevesselguide.wordpress.com/vessel-types/

Vessel PFD Symbols

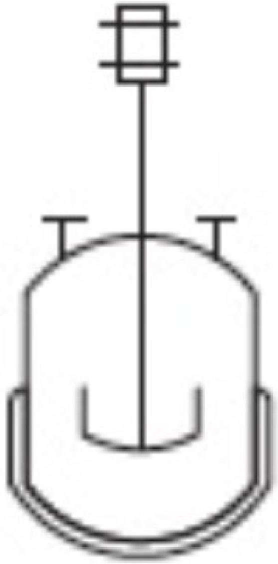


Vertical Pressure Vessel

Horizontal Pressure Vessel



Spherical Pressure Vessel



Jacketed vessel with a mixer (autoclave)

Symbol Source: Gavin’s Textbook

Vessel Design

Additional Design Considerations

- Construction material
- Vessel dimensions (wall thickness)
- Type of vessel head needed
- Openings/connections needed
- Heating specifications (jackets, coils, etc.)
- Internal fitting specifications
- Welding specifications

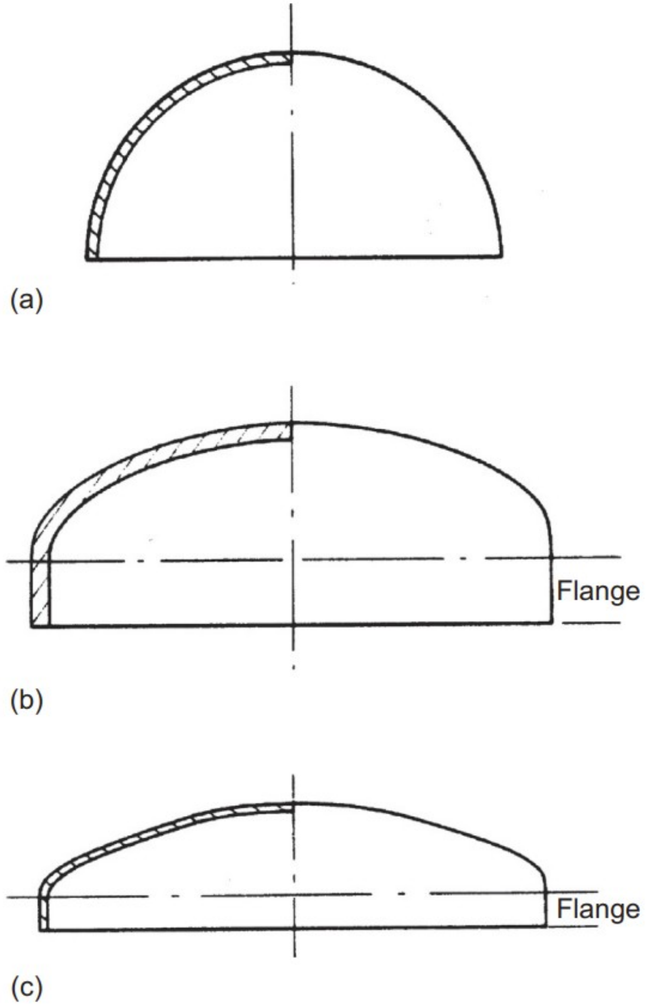
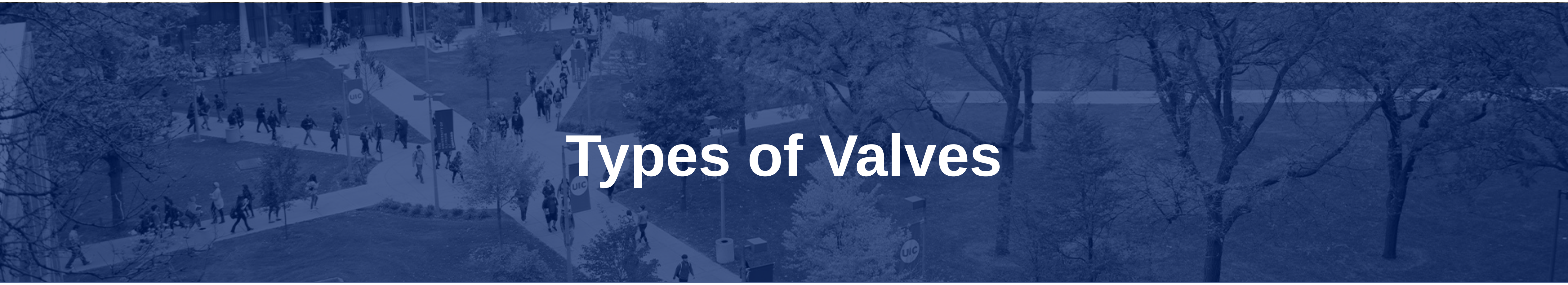


FIG. 14.5 Domed heads. (a) Hemispherical. (b) Ellipsoidal. (c) Torispherical.

Image Source: Gavin's Textbook



Types of Valves

Types of Valves

- **General Valve:** General Valve are used to control the flow of streams flow by a plug that is moved to block or allow flow through the valve. Used in any general application where streams must be shut on or off such as feed streams.
- **Three-way Valve:** Three-way Valves are used to combine streams and to control the flows of the combined stream. Three-way valves can combine and shut off streams through a plug that is controlled by the operator. Three-way are usually used to alternate streams in processes such as pumps and bypass.
- **Globe Valve:** Globe valves are used to regulate the flow rate of the stream. These valves are controlled by a pin that is very precise in regulating stream flows. Globe valves are used in precise adjustment such as steam heating.
- **Diaphragm Valve:** Diaphragm Valve uses a membrane to regulate flow rate in a valve. In Diaphragm Valves the internal membrane in the valve functions the same way as a plug in a general valve. Diaphragm Valves are mainly used in delicate process such as pharmaceuticals and food industry..

Types of Valves (Cont.)

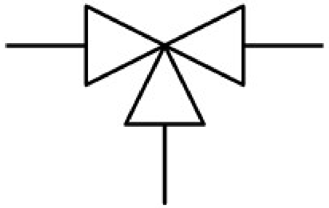
- **Pressure Relief Valve:** Pressure Relief Valves are used as safety measures to limit and control pressure inside pipes through relief port that allows excess pressure from gas or liquid to leave. Pressure Relief Valves are primary used high-pressure environments such as Gas and Oil industry.

- **Gate Valve:** Gate Valves contain a adjustable barriers that can completely shut off flows. Gate valves are known for their versatile use in many industries for complete shut offs.

- **Stop-Cock Valve:** Stop-Cock Valve are valves designed with an internal mechanical device that are used to completely stop the flow when activated. Stop-Cock Valves are primary used in heavily controlled environments such pharmaceuticals or laboratories where timed complete shut off are needed in reactions.

Valve PFD Symbols and Images

General Valve



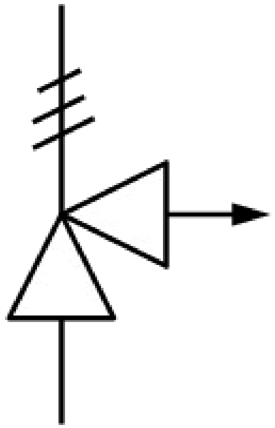
Three-Way Valve

Globe Valve

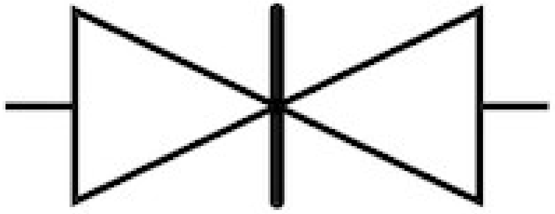
Diaphragm Valve

PFD symbols were either sourced from Gavin’s textbook or drawn by hand. Images are from: Reference sources 3,4,5,6.

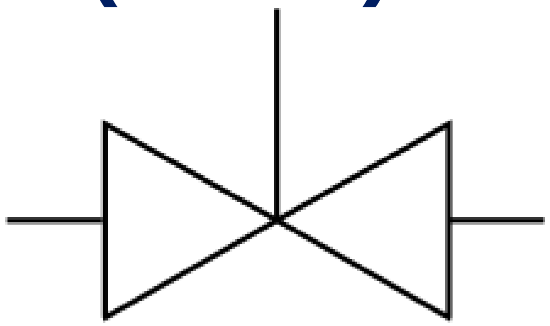
Valve PFD Symbols and Images (Cont.)



Pressure Relief Valve



Gate Valve



Stop-Cock Valve

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Considerations and Applications of Valves.

Regulating Flow and Pressure:

Pros:

- Needed for more precises and delicate applications.
- Can be used with multiple streams.
- Automated and centralized.

Cons:

- Expensive and complicated to install.
- Requires constant maintained during prolong use.

Types of Valves:

- **Pressure Relief Valve**
- **Stop-Cock Valve**
- **Diaphragm Valve**

Activating and Deactivating Streams:

Pros:

- Simple and easy to install.
- Cheap and has multiple variations of valve types.
- Low maintenance.

Cons:

- Manually operated.
- Little space required.

Types of Valves:

- **Gate Valve**
- **Three-way Valve**
- **General Valve**
- **Global Valve**

Conclusion

- The equipment used in chemical processes is necessary in the safe, efficient, and reliable functioning of the plant activities.
- The categories (heat exchangers, reactors, pumps, separators, solid handling systems and storage units) have distinct roles in the transformation and control of materials.
- their functions, symbols and design basics can aid engineers in better decisions of their processes.
- The selection of appropriate equipment will guarantee productivity, safety, and sustainability in all chemical plants.

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